

Time Pressure and Reward Incentives Shape Confidence and Accuracy in Decision-Making

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Abstract (Written by: M. Geraghty)

This study examines how time constraints and reward incentives influence decision-making accuracy and confidence. We hypothesized that participants under timed conditions would show lower accuracy and confidence compared to participants with unlimited time conditions. We also hypothesized that the addition of reward incentives would increase confidence in participants, but may also encourage risk-taking. The study involved asking college students (age 20-22) to watch a short video and answer questions about its content using a 6-point confidence scale. The study used a 2x2 design, which manipulated time limits (timed vs. untimed) and reward incentives (present vs. absent). Performance was measured based on accuracy and confidence ratings, with scores weighted based on participants' levels of certainty. We expected the time-limited group to demonstrate reduced accuracy and confidence due to reliance on faster, less analytical decision-making, while the presence of rewards was expected to enhance overall confidence in participants. These findings would support models of the speed-accuracy tradeoff and illustrate how external pressures shape decision-making processes. These results are relevant to real-world settings, such as testing environments, where individuals must make accurate decisions under time constraints.

Introduction (Written by: G. Plummer)

Many cognitive psychology studies emphasize decision-making under time restrictions, particularly when exploring the trade-off between speed and accuracy. It is believed that this tradeoff relates to people's tendency to sacrifice accuracy when making hasty decisions, whereas accuracy often improves with more time to study the provided data (Ratcliff & McKoon, 2008). This relationship is especially important in situations where people must measure their confidence in their answers while also rapidly evaluating data. Analyzing how time limitations affect both confidence and

performance can provide valuable insights into how individuals make decisions in real-world contexts where accuracy and speed must be balanced.

Due to the fact that people typically rely on their perceived certainty to evaluate their responses, confidence is essential for decision-making. Previous research suggests that confidence judgments influence how people adjust their decision-making processes over time and are inextricably linked to cognitive processing (Desender et al. 2019). When faced with time constraints, people may rely more on heuristic decision-making, thereby lowering confidence and accuracy. Reward incentives may also influence decision-making behavior by encouraging participants to commit more confidently to their responses and altering their decision thresholds (Bogacz et al., 2006). Therefore, incentives (rewards) may influence participants' confidence, even if accuracy doesn't change, by motivating them to prioritize responses they are more certain about (Koriat, 2012).

In this study, we will evaluate the effects of time constraints and reward incentives on decision-making performance and confidence. Specifically, we examine individuals' behavior in timed and untimed settings to see if time pressure affects confidence and accuracy, and whether reward incentives influence participants' willingness to be more confident in their responses. We use a 2x2 factorial design with varying reward incentives (reward vs. no reward) and time pressure (timed vs. untimed) during a confidence-based decision task. Prior research suggests that individuals adjust their decision thresholds based on work demands and time availability (Ratcliff & McKoon, 2008; Larson & Hawkins, 2023). That being said, reward incentives could encourage higher risk-taking because participants may pick more confident responses in order to maximize potential benefits (Koriat, 2012). Overall, we expect the highest levels of accuracy and confidence from participants in the untimed condition with reward incentives. We hypothesize that shorter time limits will cause participants to have less confidence and report lower accuracy, whereas with longer or unlimited time, we expect participants to have more confident and accurate answers. Additionally, when a reward is present, we expect an overall increase in confidence as participants have an incentive to commit more strongly to their responses.

Methods (Written by: B. Wu)

Experimental Design (Written by: B. Wu)

The study utilizes a 2x2 between-subjects factorial design. Participants are randomly assigned to one of four groups based on two independent variables: time constraint and scoring incentive. For the time constraint variable, participants either have unlimited time to respond or are restricted to a 15-second limit per question. For the scoring incentive variable, participants are either explicitly informed that high-accuracy, high-confidence responses qualify them for a reward, or they receive no such prompt. The primary dependent variables are the confidence-weighted accuracy score and the response time per question.

Participants (Written by: B. Wu)

We aim to recruit approximately 60 college student participants aged 20–22 ($n \approx 15$ per condition). This sample size was selected to ensure sufficient power to detect medium-sized effects (Cohen's $d = 0.5$) in a factorial design, aligning with previous research on speed-accuracy tradeoffs and confidence (Desender et al., 2019).

Materials and Apparatus (Written by: B. Wu)

The experiment is administered via a custom web application developed with coding assistance from Google Gemini. Data collection and storage are managed securely using Google Sheets via a Google Apps Script web endpoint.

The primary stimulus is a 4-minute and 12-second YouTube video. The video is embedded into the experimental interface using the YouTube IFrame API. To maintain strict experimental control over the viewing experience, all native playback controls and keyboard shortcuts are disabled, and a transparent glass CSS overlay prevents participants from clicking or interacting with the video player. Following the video, participants are evaluated using 10 True/False statements regarding the video's contents.

Procedure (Written by: B. Wu)

Participants access the experiment via unique, dynamically generated URLs that encode their specific condition assignment. Upon starting, participants provide basic demographic information, including their age, major, and ethnicity.

Next, participants review the instructions. In the reward condition, the instructions explicitly state: "You will get a final score based on both your accuracy and your confidence level. Completing this accurately qualifies you for a reward". Participants in the no-reward condition do not see this notice.

Participants then initiate the video. They are warned in advance that they cannot pause, rewind, or rewatch the footage. Immediately after the video concludes, the interface automatically advances to the testing phase. Participants are presented with the 10 statements sequentially and must rate their belief in each statement's truthfulness using a 6-point certainty scale: Definitely True, Probably True, Might be True, Might be False, Probably False, and Definitely False.

Participants in the time-constrained condition view a visible countdown timer and have exactly 15 seconds to submit an answer before the system automatically records a "No Answer" and advances to the next trial.

Scoring and Data Analysis (Written by: B. Wu)

Responses are scored using a confidence-weighted system ranging from 0 to 5 points per trial. A completely accurate, high-confidence response (e.g., selecting "Definitely True" for a factual statement) yields 5 points. Conversely, high-confidence incorrect responses yield 0 points, with intermediate confidence levels scaling proportionally (e.g., 4, 3, 2, or 1 point).

Participants do not receive trial-by-trial feedback; instead, a total cumulative score (out of a maximum 50 points) is presented upon completion of the experiment. The application automatically records the final score, the specific answers chosen, and the precise response time for every question (calculated in seconds to two decimal places).

Expected Results (Written By: C. Metoxen and M. Geraghty)

For our expected results, we expect an outcome consistent with our hypothesis: the speed-accuracy tradeoff will result in decreased confidence and accuracy in the experimental group with a time limit. It's been shown that memory recollection while stressed led to incomplete recall compared to when the stress-causing factors are absent (Lupien et al, 2007). With this in mind, we expect a similar decrease in performance in our experiment due to the time-induced stress of the participants in the experimental group. To measure our data, we will run a two-tailed t-test in order to measure if this time limit will result in a noticeably different confidence among participants. A significance threshold of a p-value of less than 0.05 would show that there is strong evidence that proves our hypothesis.

Consistent with Desender et al. (2019), confidence assessments are expected to be linked to decision-making processes and will predict how people would adjust their tactics in future decisions. It is expected that participants in the time-limited group will be more uncertain, resulting in lower accuracy and confidence ratings. Additionally, Larson and Hawkins (2023) offered more evidence that time constraints alter perceptual processing and decision thresholds, supporting our expectation that people with less time may employ heuristic decision-making approaches more often.

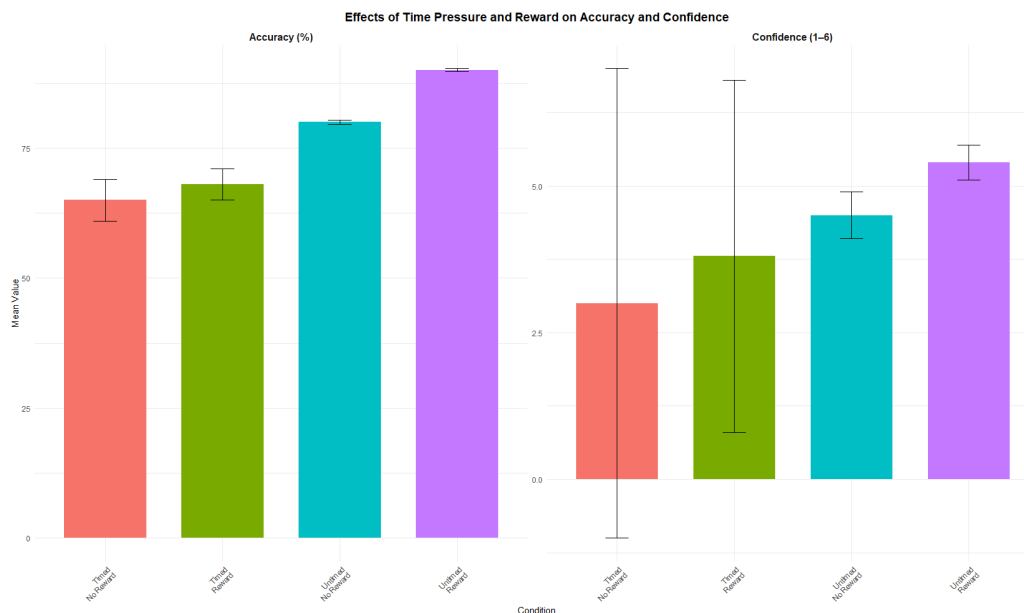


Figure 1: What we would expect our data to look like. For each of the four conditions (timed/no reward, timed/reward, untimed/no reward, and untimed/reward), the average accuracy (left) and confidence ratings (right) are shown. While reward boosts confidence and somewhat improves accuracy, time pressure decreases both accuracy and confidence. The untimed reward condition exhibits the best performance. The standard error of the mean is shown by error bars.

Results: (Written By: C. Metoxen)

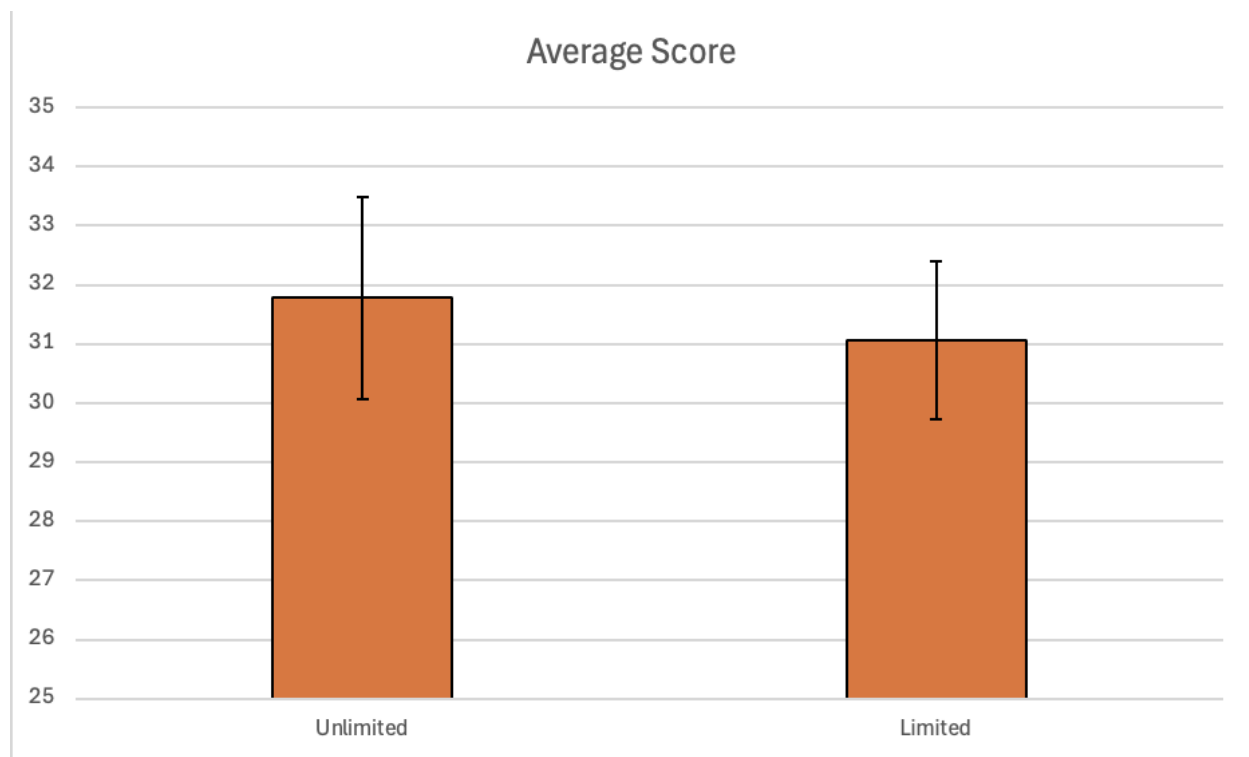


Figure 2: Results from our data on the average score of the two groups. Standard error of Mean for the experimental, limited time group is ± 1.33 , and the Control, Unlimited time group is ± 1.71 . Results of the two-tailed T-test gave the following values. The mean difference between these two groups is 0.72. The standard error of the difference is 2.17. The p-value is 0.743. The degrees of freedom of this data is 17.67. The test statistic is 0.333. The 95% confidence interval is $[-3.83, 5.28]$. With the p-value being greater than 0.05, we fail to reject the null hypothesis. We are 95% confident the true difference between the mean of these groups falls between $[-3.83, 5.28]$, and because this interval contains 0, it confirms the difference in data is not statistically significant.

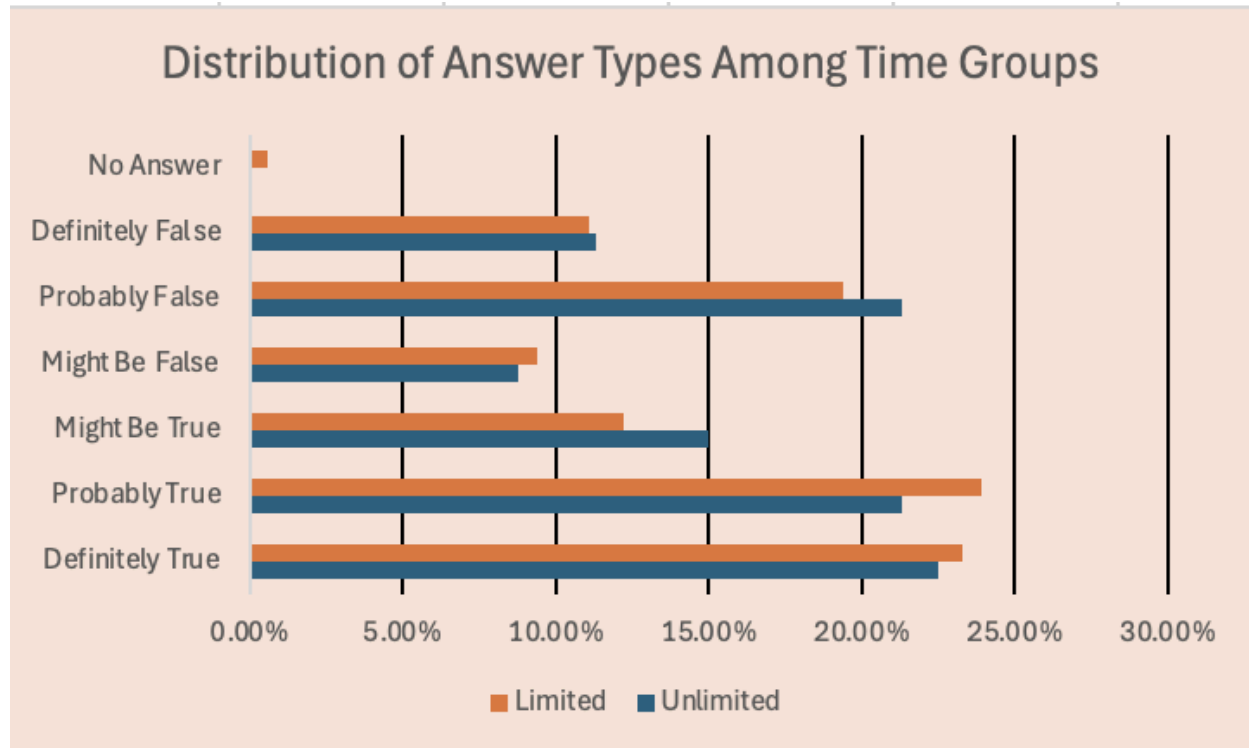


Figure 3: A graph showing the raw distribution of answer types, and overall confidence, of the two study groups. All differences in the distributions are minimal, showing marginal difference between the two groups.

Neuroscience Extensions (Written by: M. Geraghty)

Our behavioral paradigm could be extended by including the use of electroencephalography (EEG) in the experimental design. EEG could be incorporated to measure brain activity while participants complete the experimental task under time-limited and unlimited conditions. This would be a useful recording method because EEG captures rapid changes in neural activity, which would enhance the experiment's study of decision-making. Prior research has demonstrated that a centroparietal EEG signal tracks how evidence accumulates and the confidence of the participants prior to making a decision. So, in untimed conditions, we would anticipate more significant centroparietal accumulation signals, which would indicate improved evidence integration before decision commitment (Herding et al., 2019). Additionally, we would also predict increased activity in frontal brain regions, such as the anterior cingulate cortex (ACC), which is involved in weighing options and adjusting behavior during decision-making

(Walton et al., 2007). Based on our hypothesis, the time-limited group would likely show weaker accumulation signals and more activity in the frontal brain regions, specifically the ACC. This would suggest that the individuals in the time-limited group are making faster decisions with less evidence accumulation and more pressure due to the time constraint.

Discussion (Written By: C. Metoxen)

Several factors may influence the outcome and interpretation of the expected results. Our results could be skewed due to the absence of difficult questions that led to faster responses as a result of the limited time, and being as accurate as the control group without the time limit. The Drift-Diffusion Model (DDM) is a model used for assessing the speed-accuracy tradeoff. It assesses a person's evidence accumulation through noise before eventually reaching a threshold of either one choice or another. In our project, we these representative thresholds would either be “correct” or “incorrect”, with our hypothesis being that with the noise of a time limit being present, evidence accumulation would not be sufficient to reach these thresholds, and therefore participants would guess for answers, leading to greater disparity in both confidence in answers as well as accuracy for the time limited experimental group versus that of the unlimited time, control group.

However, DDM modeling, including the speed-accuracy tradeoff, does not accurately account for test questions in the real world (Fudenberg et al., 2018). The DDM assumes participants already know the true value difference, with reducing perceptual noise being the barrier to reaching a threshold. In a revised model, participants would be uncertain about the values of the thresholds themselves, and time would be used to learn what a correct answer might be. Relating this information to our project, under the DDM model, taking more time to answer questions would lead to more accurate answers. Through the revised model, taking more time would not guarantee more accurate answers, as harder questions would lead to more time taken, thus leading to more inaccuracy and lower confidence. This model would also explain why, with simple questions, accuracy and confidence were as high between both groups despite one group having a capped time limit. As a result, if our questions were more

complex, we would expect a greater disparity between the two groups, leading to greater support of our hypothesis.

There are multiple limitations presented in this study that may have resulted in the difference between true data and our actual data; one of these limitations may have been sample size. In our study, we were only able to get 26 total participants, with 18 participants in the limited time, experimental group, and 9 in the unlimited time, control group. This sample size could have resulted in skewed data, as with only 9 participants in the control group, data might not be representative of the target population of our study, college students aged 20-22. This limited sample size may have produced a type II error, or limited data showing a result that showed no correlation between variables, despite one potentially being present with a greater sample size. With a repeated study with more participants, it is possible for our hypothesis to be substantiated.

One final limitation may have been the motivation of the participants and its effect on performance in our study. This may not have been motivating enough, as it is predicted that with higher motivation, higher accuracy and therefore confidence would result, as well as improved answer times. Another justification of how motivation could have affected this study is through the pressure effect. Motivation can improve neural processing as well as be effective at filtering out noise, leading to positive outcomes in test-like environments (Ranger et al. 2021). Relating this to our results, our time-limited experimental group may have felt motivated through the pressure effect, or due to the increased urgency, this group may have had increased focus when answering questions. The unlimited time control group did not have this pressure effect, therefore may not have been as motivated. With our hypothesis being that the speed-accuracy tradeoff would result in the experimental group being less confident and accurate, placing the time constraint may have led to motivation in performance, which may have resulted in the small disparity of results between the two groups.

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Figure 3: Cal

References: Grace, Mikaela, Cal